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Roll No

MA-220(CE)-CBCS

B.E., IV Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Mathematics - III

Time : Three Hours

Maximum Marks : 60

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Find a series of sines and cosines of multiples of x , which will represent $x + x^2$ in the interval $-\pi < x < \pi$. Hence

show that $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

- b) Expand $f(x) = \pi x - x^2$, $0 < x < \pi$ in a half-range sine series.

2. a) Find the Fourier transform of $f(x) = \begin{cases} x, & |x| \leq a \\ 0, & |x| > a \end{cases}$

- b) Find the Fourier cosine transform of

$$f(x) = \begin{cases} \cos x, & 0 < x < 9 \\ 0, & x > 9 \end{cases}$$

3. a) Find $L\left\{\frac{\sin t}{t}\right\}$

- b) Prove that $L\left\{\frac{\cos \sqrt{t}}{\sqrt{t}}\right\} = \frac{\pi}{s} e^{-1/4s}$

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4. a) Find $L^{-1}\left\{\frac{s+1}{s^2+6s+25}\right\}$
- b) Solve $y'' - 2y' + 2y = 0$, given $y(0) = y'(1) = 1$
(By Laplace transform method)
5. a) Use Cauchy-Riemann equation to find v when
 $u = 3x^2y - y^3$.
- b) Evaluate the integral $\int_0^{1+i} z^2 dz$
6. a) Evaluate the following integral using Cauchy's integral formula:
- $$\int \frac{(4-32)}{z(z-1)(z-2)} dz$$
- b) Show that $\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta} = \frac{2\pi}{\sqrt{a^2-b^2}}$, where $a > b > 0$.
7. a) Find a real root of the equation $f(x) = x^3 - 4x - 9 = 0$,
using bisection method in four stages.
- b) Apply False position method to solve the equation
 $3x - \cos x - 1 = 0$
8. a) By using Newton-Raphson method find the root of
 $x^4 - x - 10 = 0$, which is nearer to 2, correct to three
places of decimal.
- b) Find a quadratic factor of the polynomial
 $x^4 + 5x^3 + 3x^2 - 5x - 9 = 0$, starting with $p_0 = 3, q_0 = -5$,
by using Bairstow's method.
